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ORIGINAL PAPER

Detecting caste and migration hate speech in low-resource Tamil language

Bharathi Raja Chakravarthi[1,3] · Saranya Rajiakodi[2] · Rahul Ponnusamy[3] · Bhuvaneswari Sivagnanam[2] · Sara Yogesh Thakare[4] · Sathiyaraj Thangasamy[5]

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Abstract
The Indian constitution categorizes its population into groups such as Scheduled Castes, Scheduled Tribes, Other Backward Classes, and Forward Castes, reflecting historical inequalities that influence social dynamics and discrimination. Migrants who are relocating within the country for better opportunities are often viewed as outsiders, leading to concerns about job security and crime, fostering hate and discrimination against them. Social media has exacerbated these issues, becoming hotspots for caste and migration-related hate speech, especially in low-resource languages. This study introduces a novel dataset specifically curated to detect hate speech related to caste and migration in the low-resourced Tamil language. Using this dataset, we benchmarked the dataset with baseline experiments with the highest macro-F1 score of 0.73. We also created custom-modified models integrating custom loss functions, adapter-based fine-tuning, and parameter-efficient fine-tuning techniques. To support further research, we released the dataset, conducted a shared task, and ranked participant systems. By publicly releasing this dataset, we aimed to facilitate further study and improve the detection and mitigation of hateful content related to caste and migration on social media.

Keywords Caste and migration hate speech · Low-resourced dataset creation · Large language models · Model optimization · Adapters · Parameter-efficient fine-tuning · Tamil

1 Introduction

The Indian constitution divides the people into four main social groups: Scheduled Castes (SC), Scheduled Tribes (ST), Other Backward Classes (OBC), and the General/Forward castes (FC). Title 341(1) says that "Scheduled Castes" are certain castes, races, tribes, or subgroups within castes or races (Kijima, 2006). The Constitution of India says that Scheduled Tribes are tribes or tribal communities,

Extended author information available on the last page of the article

or certain parts or groups within the tribes or tribal communities, that are publicly recognized as those under Article 342 of the Constitution. According to the Mandal Commission (1978), people who are not in the Scheduled Castes or Scheduled Tribes groups are in the Other Backward Classes group. Other Backward Classes group have been pushed to the edges of society in the past, and they still face social, economic, and educational inequality. In addition, "forward castes" (FC) are social groups that don't fit into any of the three groups listed above (Mal & Saikia, 2024; Rawal & Swaminathan, 2011; Shukla et al., 2024). The caste system, which has been prevalent in India for many years, is a societal framework deeply rooted in religious and cultural norms. This system divides the population into a hierarchical structure of distinct classes (Binny, 2022). Caste is a very important factor in how people's social and economic lives turn out in modern India. Unfairness is caused by a structured system, not just cultural identity, that decides who has benefits and disadvantages (Mosse, 2018).

Policymakers, governments, civil society groups, and experts working on local and European projects to track, stop, and fight the spread of casteism and xenophobia in our Indian society all have different ideas about how to solve these problems (Abubakar et al., 2022). At the moment, experts from a number of different fields are focusing on mass media and different aspects of the language used in these settings to find contrasting hate speech (Cuzzocrea et al., 2025). Data network analysis, social psychology, and discourse analysis are all fields that study and keep an eye on them (Arce-García et al., 2024). However, the problems they create make it difficult for artificial intelligence and computational linguistics to handle. It has been one of the hottest topics in Natural Language Processing (NLP), text classification, and opinion mining over the last few years to find hateful material on social media (Basile et al., 2019; Bosco et al., 2018).

Social media has become an integral part of our daily lives, which transforms the way we communicate, connect, and disseminate information (Chakravarthi et al., 2023a). Such a cyberworld enables users to share their viewpoints and articulate their opinions globally. It substantially impacts society, allowing us to influence everything around us. Nowadays, hate speech concerning sensitive topics are proliferating in social media platforms such as X (Twitter), YouTube and Facebook (Shanmugavelan, 2022). Globally, the detection of hate speech text across various realms on social media has become a pivotal field due to the growing prevalence of online hate speech and its likelihood to trigger real-world violence and discrimination (Rodríguez-Peral et al., 2025). Researchers across the world have developed various methodologies, including Machine Learning (ML), Deep Learning (DL) and NLP techniques, to identify and mitigate hate speech. However, the challenge lies in the nuances of language and cultural aspects of diverse regions across the world (Abubakar et al., 2022). In the context of Indian scenarios, the issue of caste and migration-related hate speech is particularly increasing on social media platforms because of India's cultural heritage and historical caste-based discrimination (Banaji & Bhat, 2022).

Social media platforms have become hotbeds for people to comment on their opinions related to caste-based prejudices and anti-migrants, which typically leads to social unrest (Fouskas et al., 2022; Udupa, 2020). As India has diverse

linguistics, Indian researchers have been developing algorithms to detect and flag content posted in native languages depending on the contextual understanding of caste and migrated-related derogatory comments (Pohjonen & Udupa, 2017). In states such as Tamil Nadu, located in the southern part of India, the caste systems are integrated in social and cultural contexts, particularly in rural areas (Rajiakodi et al., 2024). Even in this contemporary society in Tamil Nadu, the caste system periodically determines the different spheres of people's lives, such as marital bonds, political manipulation, social interactions, and social status (Vaghela et al., 2022). People are using social media to spread hate speech against a particular caste, which intensifies the tension and even worsens the social divides. Offenders are exploiting anonymity to post hateful or inflammatory content by demeaning a particular caste without facing repercussions. In addition, such hate speech typically goes viral on social media and it easily reaches a vast audience, especially youth, so the digital spread of hate speech can lead to discrimination and further marginalization of vulnerable groups. Such hate speech not only affects the targeted caste but also facilitates an environment of anxiety and hostility (Kirasur & Jhaver, 2024).

In addition, another growing issue in Tamil Nadu in social media is hate speech targeting migrants, particularly with respect to interstate migrates from northern and eastern states of India like Bihar, Uttar Pradesh, West Bengal, Assam, and also from neighboring states who have moved to Tamil Nadu for employment and a better-quality life (Rawat et al., 2024). Hate speech comments regarding migrants often involve blaming them for crime, accusing them of taking jobs away from local people, and even shaming their physical appearance. To address these challenges, a significant demand for NLP models is required that can effectively handle the complexities of Tamil and code-mixed texts related to caste/migration hate speech on social media.

Our research aims to bridge this gap by enhancing the development and implementation of advanced NLP techniques which are specially used to detect and classify caste/migration hate speech in Tamil and code-mixed (Tamil-English) texts on social media, as well as distinguishing it from not-caste/migration-related content, thereby it mitigates the spread of harmful content and promotes a more harmonious digital society.

Research Question: How can NLP models be effectively optimized and utilized to detect and classify caste/migration-related hate speech in Tamil and code-mixed texts on social media platforms, considering its linguistic and cultural features?

In this paper, we have discussed the key contributions of our work. They are as follows:

1. We have curated the first ever novel dataset for detecting the hate speech related to caste/migration in the Tamil language and made it publically available for future research.
2. We provided the evaluation framework for this dataset and performed baseline experiments using various DL architectures.

3. We conducted experiments with the custom model by incorporating custom loss functions, Adapters, P-tuning and Low-Rank Adaptation (LoRA) techniques to improve its performance.
4. We performed an error analysis on the optimal model to identify its limitations and highlight potential areas for future enhancements.
5. We organized a shared task for researcher to measure various method's performance to the dataset.

2 Related work

The identification of hate speech has emerged as a significant issue in the age of social media and digital communication platforms. The use of offensive language is connected to a range of linguistic and social problems, including the use of abusive and violent language, cyberbullying, racism, extremism, radicalization, toxicity, profanity, and hate speech. The surge in hate speeches has resulted in a growth in the quantity of corpora that annotate abusive languages. Zampieri et al. introduced a novel dataset in a study. This dataset provides a systematic structure for classifying offensive language according to its type and intended recipient. Jeong et al. have recently presented a Korean abusive language dataset in their concurrent work. The dataset contains annotations for the target group, offensive span, target span, and level of offensiveness. They focus on justifying the decision for offensiveness by using additional activities, such as identifying the target of the insult and assessing the severity of the offense. The objective of this study is to enhance the accuracy of offensive content classification by integrating additional tasks such as Automatic Lexical Detection (ALD) and Sentiment Analysis (SA).

Researchers have been investigating many methods, such as machine learning, deep learning, and transformer-based models, to effectively address the increase in hate speech events (Fortuna & Nunes, 2018). The majority of study focuses on commonly spoken languages, such as English (Davidson et al., 2017; Waseem & Hovy, 2016). Nevertheless, the lack of a substantial corpus for Indian languages has impeded the advancement of NLP models, resulting in their limited exploration (Chakravarthi et al., 2020; Ponnusamy et al., 2024). Transformer models for several Indian languages, including Hindi, Marathi, and Bengali, have been recently made available (Chhaya et al., 2024). The findings also indicate that models trained on monolingual datasets exhibit superior performance. Several significant contests, like SemEval 2018, SemEval 2019, HASOC 2020, and GermEval 2018, have been held to explore enhanced outcomes for NLP jobs. Researchers have collected datasets from many sources for languages other than English. This has facilitated the investigation of different methods and feature sets for detecting hate speech, such as machine learning techniques (Kumaresan et al., 2023; Sreelakshmi et al., 2024; Subramanian et al., 2022).

Recent advancements in deep learning methodologies have underscored the necessity of massive amounts of labeled data to achieve high performance (Brown et al., 2020; Liu et al., 2019). However, the majority of research in this area has predominantly focused on English and a few other major languages, leaving many other

languages underrepresented and lacking in sufficient labeled datasets for training robust models (Bender et al., 2021; Hedderich et al., 2021). This disparity presents a significant challenge, particularly for low-resource languages, which often lack the volume of undesirable data needed for effective model training (Joshi et al., 2020; Nekoto et al., 2020).

Many low-resource languages have attempted to bridge this gap by developing their own language models. These models are typically created by collecting open-source data available online and pretraining it to suit the specific linguistic features and usage of the language (Kakwani et al., 2020; Joshi et al., 2020). Subsequently, these pretrained models are fine-tuned for various specific tasks such as offensive language detection, hate speech identification, emotion detection, and translation (Hedderich et al., 2021; Ruder et al., 2019). Notably, Tamil is one such language where efforts have been made to develop tailored models to address its unique linguistic needs (Chakravarthi et al., 2021, 2024; Ponnusamy et al., 2024; Premjith et al., 2024; Ravikiran et al., 2024). The advent of transformer-based models has significantly transformed the field of NLP. These models have demonstrated superior performance across a range of tasks but come with the drawback of being resource-intensive, particularly during the fine-tuning process (Devlin et al., 2019; Vaswani et al., 2017). Fine-tuning transformers for specific tasks can be prohibitively costly in terms of computational resources and time, which is a critical issue for many researchers and practitioners working with low-resource languages (Kaplan et al., 2020; Raffel et al., 2020). Prior studies (Chakravarthi et al., 2021, 2023b; Kumaresan et al., 2024; Subramanian et al., 2022; Waseem et al., 2017) have categorized offensive occurrences into two distinct groups: explicit and implicit. However, none of the aforementioned studies specifically tackle the problem of hate speech on caste and migration in the Indian context Tamil. This study is one of the few that addresses the problem of Tamil language caste hate by incorporating additional tasks that are important.

To mitigate these challenges, various methods have been introduced to optimize the fine-tuning process, making it more resource-efficient. One prominent approach is Adapter-based fine-tuning, which involves adding lightweight adapter modules to the pretrained models. These adapters can be fine-tuned for specific tasks while keeping the majority of the model's parameters fixed, thus significantly reducing the computational cost (Houlsby et al., 2019). Additionally, Parameter-Efficient Fine-Tuning (PEFT) techniques, such as LoRA, P-tuning, and Prompt tuning, have been developed. These techniques allow for effective fine-tuning by optimizing a smaller subset of parameters, thereby making the process more efficient and accessible (Hu et al., 2022; Lester et al., 2021; Liu et al., 2022).

These advancements are particularly relevant for detecting hate speech and other forms of offensive content in low-resource languages like Tamil. By leveraging optimized fine-tuning techniques, researchers can develop models that are both effective and resource-efficient, enabling broader application and deployment in real-world settings. This is crucial for addressing the nuanced and culturally specific forms of hate speech that are prevalent on social media platforms, especially in linguistically diverse regions such as India (Chakravarthi et al., 2021; Subramanian et al., 2022). In our work, we have created a standard dataset for the automatic detection of hate

speech on caste or migrants in Tamil language. We experimented with various technique, starting from traditional fine-tuning to recent PEFT methods with the adapters. The detailed explanation of the dataset creation and the performance of all the experiments conducted are clearly discussed in the upcoming sections.

3 Dataset

The caste/migration hate speech detection dataset is one of its kind which focuses on identifying the hateful/abusive/derogatory comments directed against certain castes or migrated people in Tamil Nadu. It is collected as a part of ongoing research on developing a model for automatically detecting harmful content towards a certain community of people. Because of the increasing number of abusive comments on social media platforms, a need for this dataset in NLP for low-resource languages is crucial, especially for Tamil. So this dataset plays a crucial role in detecting abusive and derogatory comments. It also focuses on various forms of hate speech and provides unique insights into regional and cultural-specific discrimination. The workflow of the dataset design is, we start with the data collection process, then proceed with data preprocessing, annotation of the dataset and splitting the dataset into training, testing, and validation of the dataset for our task.

3.1 Collection

For creating a caste/migration hate speech detection dataset, we need to collect the good quality data. For this step, the dataset has been carefully curated from the YouTube platform by gathering comments from videos discussing caste and migration-related issues. The reason why the YouTube platform has been chosen is for its diverse social interactions so that we can get various kinds of public opinion and social debates. It is the first ever dataset that has been solely focusing on caste and migration hate speech in Tamil language.

To collect the comments from the YouTube videos, we have used one of the Python libraries named "youtube-comment-scraper".[1] This tool has scraped the YouTube comments of the given videos including the replies of these comments and these replies have been considered as separate comments. The videos have been chosen based on hashtags and by manual key searches. The below illustrated word cloud representation that highlights various hashtags, including 'vadakan' (North Indian), 'devar' (caste) and more shown in Fig. 1 and manual keywords such as 'Agamudiyar' (caste), 'Melpathi temple issue' (a particular caste named devar) and others shown in Fig. 2 which are in both Tamil and code-mixed (Tamil-English). The words illustrated in the word clouds are the most frequently used keywords to choose the YouTube videos. Figure 1 listed the hashtags used for the search, whereas

[1] https://github.com/bellingcat/youtube-comment-scraper.

Fig. 2 describes the keywords related to certain events associated with the debate of the issues pertaining to caste/migration.

3.2 Data preparation

The second step in the dataset design is data pre-processing or data preparation. In this step, we proceeded to remove unrelated content from the collected YouTube comments like removing usernames, removing URLs and removing sentences that contained less than three words. Figure 3 describes the most frequently used words in the dataset. After collecting a total of 10,173 comments, we retained 7875 for annotation following the preprocessing step.

3.3 Annotation

The crucial task of constructing a dataset is annotation. The annotation process for this task involves three annotators who will be manually annotating a total of 7875 comments. These human experts will annotate the comments into two different categories, such as Hate speech on caste/migration which has been labeled as 1 and Not-hate speech on caste/migration has been labeled as 0. Before sending the data to the annotators, we removed all kinds of personal information and privacy related data from the comments, which is done in the data pre-processing step. It is to ensure the confidentiality and privacy of the commentator.

Each comment is annotated by three annotators. If there is any disagreement in determining the label for a comment, the final decision will be based on the majority vote. Specifically, if two out of the three annotators assign the same label to the comment, that label will be adopted. For this annotation task, the annotators are all well-versed in the Tamil language. The group of heterogeneous annotators comprising individuals from various castes including historically marginalized and dominant groups, different age groups from young-adults (between 20 and 29 years old) to middle-aged (between 30 and 45 years old), gender and professional experience. Figure 4 illustrates the example of sample dataset after the meticulous annotation, and Table 2 shows the information about the annotators.

3.3.1 Annotation guidelines

To do annotation in this task, we have a set of guidelines given to the annotators. First, it is mentioned in our guidelines that there are two categories: Hate speech on caste/migration and Not-hate speech on caste/migration. Below are the definition for each category:

- **Hate speech on caste/migration:** Comments that include abusive, disrespectful, discriminatory language, as well as ridiculing and mockery, and content that aims to delegitimize certain caste or migrated individuals.

Fig. 1 This illustrated word cloud represented the hashtags used for the YouTube video search

- **Not-Hate speech on caste/migration:** Comments that do not include any discriminatory or derogatory remarks towards individuals based on their caste/migration status.

In the second step, the annotators have to read through the entire comment and make sure that the entire context is understandable. After that, they have to mark the labels. For example, specific criteria are used to determine whether the context is hate speech related to caste (Fig. 5) or migration (Fig. 6). These criteria include terms such as *vadakan*, *marvadi*, *parayan*, *vaduka pallan*, *vaellalar*, and *mukkulathor*.

3.3.2 Annotation difficulties

In this section, we will be discussing the difficulties faced by the annotators doing annotation process. The annotator finds it challenging to determine the intended context of comments that contain words related to multiple genres, such as work, play, and others. Another problem encountered during annotation is the mix of

Fig. 2 This illustrated word cloud displays the words used for the manual search of certain issues related to caste/migration from the YouTube videos

languages, such as Hindi within Tamil comments, which makes it difficult to understand the meaning of the comments. Example for annotation difficulties: Fig. 7.

3.4 Inter annotator agreement (IAA)

To determine the quality of annotated data, Statistical measures such as Cohen's kappa, Fleiss kappa, Scott's pi, or Krippendorff's alpha will be used (Lombard et al., 2010). For this task, we have chosen Krippendorff's alpha measure.

$$\alpha = 1 - D_o \div D_e \tag{1}$$

The reason for choosing Krippendorff's alpha is that it can handle various data types and missing values in case one of the annotators does not label all the items in the dataset. It can be used when there are more than two annotators, and if it achieves Krippendorff's alpha score more than 0.80, then there is a strong agreement and considered reliable. For our task, the Krippendorff's score is 0.83441.

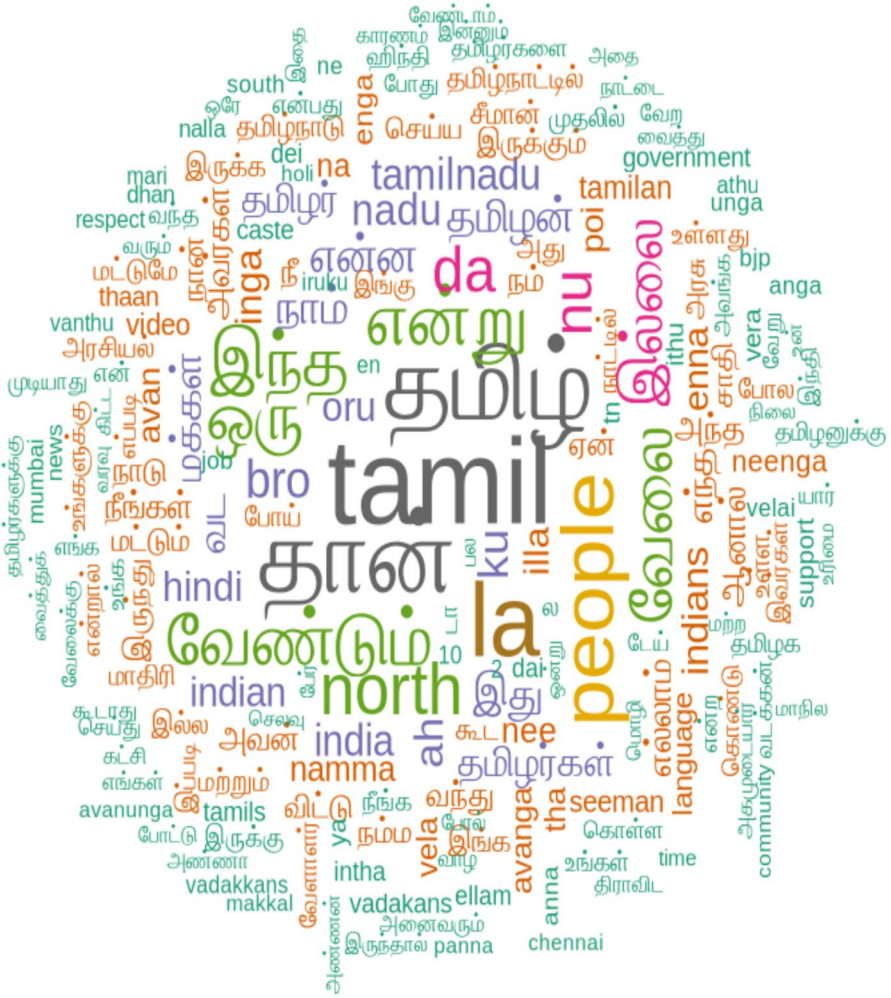


Fig. 3 This word cloud represents the visual summary of the whole dataset, highlighting the frequent terms. The word 'Tamil' appears the most in the dataset

4 Experimental setup

In our experimental setup, we leveraged a powerful computational environment to facilitate efficient processing and thorough analysis. The core hardware component was the Nvidia Tesla V100 GPU, chosen for its exceptional performance in handling intensive computational tasks. The system was equipped with 156 GiB of RAM, which ensured ample memory capacity for managing large-scale datasets and large language learning models. The GPU featured 16 GiB of dedicated memory, enabling efficient processing of graphics-intensive operations. Our operating system of choice was Ubuntu, selected for its stability and compatibility with a wide range of

machine learning frameworks. Python was utilized as the primary programming language, owing to its extensive library support and strong community. For coding and development, we employed the VS Code editor, which provided a robust and flexible environment for writing and managing code. This setup was critical in ensuring a stable and effective platform for conducting our experimental procedures. Table 3 provides an overview of the hardware and software configurations employed.

The dataset was divided into three parts, such as training, development, and testing sets, with ratios of 65, 15, and 20% to ensure comprehensive analysis and support effective model training. The accompanying Table 1 gives the breakdown of the comments distributed across these subsets. The multilingual models (obtained from Hugging Face[2]) used in these experiments are:

1. bigscience/bloom-560 m[3]
2. google/muril-base-cased[4]
3. ai4bharat/indic-bert[5]
4. l3cube-pune/tamil-bert[6]
5. l3cube-pune/tamil-sentence-bert-nli[7]
6. FacebookAI/xlm-roberta-base[8]
7. facebook/xmod-base[9]

All of these models are multilingual and are suitable for working in the Tamil language. They have been used for the downstream task of sequence classification. We performed experiments with 5 different learning rates to find each model's best possible learning rate, keeping all other parameters constant. The configurations used were as follows: weight_decay: float = 0.01. We employed the AdamW optimizer[10] with the following parameters: Beta1: 0.9, Beta2: 0.999, adam_epsilon: float = 1e-08. The training was conducted for 5 epochs, as this is the standard for training transformer models, with a batch size of 8, which optimizes memory usage and allows efficient training on GPUs. All models were implemented in the PyTorch[11] framework.

[2] https://huggingface.co/models.
[3] https://huggingface.co/bigscience/bloom-560m.
[4] https://huggingface.co/google/muril-base-cased.
[5] https://huggingface.co/ai4bharat/indic-bert.
[6] https://huggingface.co/l3cube-pune/tamil-bert.
[7] https://huggingface.co/l3cube-pune/tamil-sentence-bert-nli.
[8] https://huggingface.co/FacebookAI/xlm-roberta-base.
[9] https://huggingface.co/facebook/xmod-base.
[10] https://pytorch.org/docs/stable/generated/torch.optim.Adam.html.
[11] https://pytorch.org/.

4.1 Fine-tuning

The pretrained models utilized to address this problem were selected based on their effectiveness in processing the Tamil language and knowledge from the Tamil culture which are listed below. *BLOOM:* BLOOM (BigScience Large Open-science Open-access Multilingual Language Model) (Scao et al., 2022) is based on a Decoder-only architecture derived from Megatron-LM's GPT2. For our task we used BLOOM-560 m, it has 559,214,592 total parameters, with 256,901,120 parameters dedicated to embeddings. It operates with 24 layers, each utilizing 16 attention heads. The hidden layers within BLOOM are 1024-dimensional, and it processes sequences up to 2048 tokens in length. The model is trained using the Cross Entropy objective function with mean reduction. To train BLOOM, the ROOTS corpus (Laurençon et al., 2022) was utilized, which comprises diverse sources in 46 natural languages and 13 programming languages, totaling 59 languages in total.

IndicBERT: IndicBERT (Kakwani et al., 2020) is a multilingual ALBERT model that has been trained on extensive corpora encompassing 12 major Indian languages, totaling 9 billion tokens and occupying a storage size of 120 GB. ALBERT was selected as the foundational model due to its reduced parameter count, facilitating easier distribution and deployment in various downstream applications.

MuRIL: MuRIL (Khanuja et al., 2021) is a multilingual model based on the BERT architecture. It was pre-trained from scratch using a combination of datasets including Wikipedia, Common Crawl, PMINDIA, and Dakshina corpora, which cover 17 Indian languages. MuRIL is trained on both parallel and monolingual segments of these datasets. The model utilizes a self-supervised masked language modeling task during training to learn language representations effectively.

TamilBERT: Existing multilingual models like MuRIL, XLM-RoBERTa, and IndicBERT have been fine-tuned on monolingual corpora. This process has resulted in models specifically tailored for Hindi, termed as HindBERT, HindRoBERTa, and HindAlBERT, as well as corresponding Devanagari-specific models named DevBERT,[12] DevRoBERT,[13] and DevAlBERT.[14] These models have demonstrated excellent performance on various tasks such as classification and Named Entity Recognition (NER). Building on the success of HindBERT and DevBERT, monolingual models have been released for other Indian languages including Kannada, Telugu, Malayalam, Tamil, Gujarati, Assamese, Odia, Bengali, and Punjabi. These models are respectively referred to as Kannada-BERT,[15] Telugu-BERT,[16] Malayalam-BERT,[17] Tamil-BERT,[18] Gujarati-BERT,[19] Assamese-BERT,[20] Odia-BERT,[21]

[12] https://huggingface.co/l3cube-pune/hindi-marathi-dev-bert.
[13] https://huggingface.co/l3cube-pune/hindi-marathi-dev-roberta.
[14] https://huggingface.co/l3cube-pune/hindi-marathi-dev-albert.
[15] https://huggingface.co/l3cube-pune/kannada-bert.
[16] https://huggingface.co/l3cube-pune/telugu-bert.
[17] https://huggingface.co/l3cube-pune/malayalam-bert.
[18] https://huggingface.co/l3cube-pune/tamil-bert.
[19] https://huggingface.co/l3cube-pune/gujarati-bert.
[20] https://huggingface.co/l3cube-pune/assamese-bert.
[21] https://huggingface.co/l3cube-pune/odia-bert.

Table 1 This table shows the distribution of sentence in the dataset after annotation

Sets	Hate speech on caste/migration	Not-Hate speech on caste/migration	Total
Train	2052	3303	5355
Development	351	594	945
Test	602	973	1575
Total	3005	4870	7875

Table 2 Annotator's demographics

Gender	Qualification	Age group
Male	PhD	Middle-aged
Female	PG	Young-adults
Female	PG	Young-adults

In Age group column Young-adults denoted the annotators age is between 18 and 25.Middle-aged denoted between 26 and 45

Bengali-BERT,[22] and Punjabi-BERT.[23] For our task, we used Tamil-BERT (Joshi, 2022) for our task.

TamilSBert: Sentence-BERT (SBERT) adapts the BERT architecture and fine-tunes pre-trained BERT models specifically for generating sentence embeddings. SBERT incorporates additional training techniques aimed at enhancing the effectiveness of training sentence embeddings. The model is trained using supervised datasets such as Natural Language Inference (NLI) and Semantic Textual Similarity (STS), which aid in capturing sentence semantics. TamilSBERT (Deode et al., 2023), also known as TamilBERT, is a variant trained specifically for Tamil. It leverages the NLI dataset, which has been translated into Tamil through machine translation from the original English version. This adaptation ensures that the model learns to encode meaningful representations of Tamil sentences based on the semantics inferred from the NLI tasks.

XLM-RoBERTa: XLM-RoBERTa is a multilingual version of RoBERTa that has been pre-trained on a substantial dataset from CommonCrawl, amounting to 2.5TB of filtered data covering 100 languages (Conneau et al., 2020). This model, like RoBERTa, employs a transformer-based architecture and is pre-trained in a self-supervised manner. Specifically, XLM-RoBERTa (XLM-R) (Conneau et al., 2020) is pre-trained using the Masked Language Modeling (MLM) objective. In MLM, 15% of the words in a sentence are randomly masked, and the model is tasked with predicting these masked words based on the context provided by the entire sentence. This differs from traditional recurrent neural networks (RNNs) that process words sequentially or autoregressive models like GPT that internally mask future tokens.

[22] https://huggingface.co/l3cube-pune/bengali-bert.
[23] https://huggingface.co/l3cube-pune/punjabi-bert.

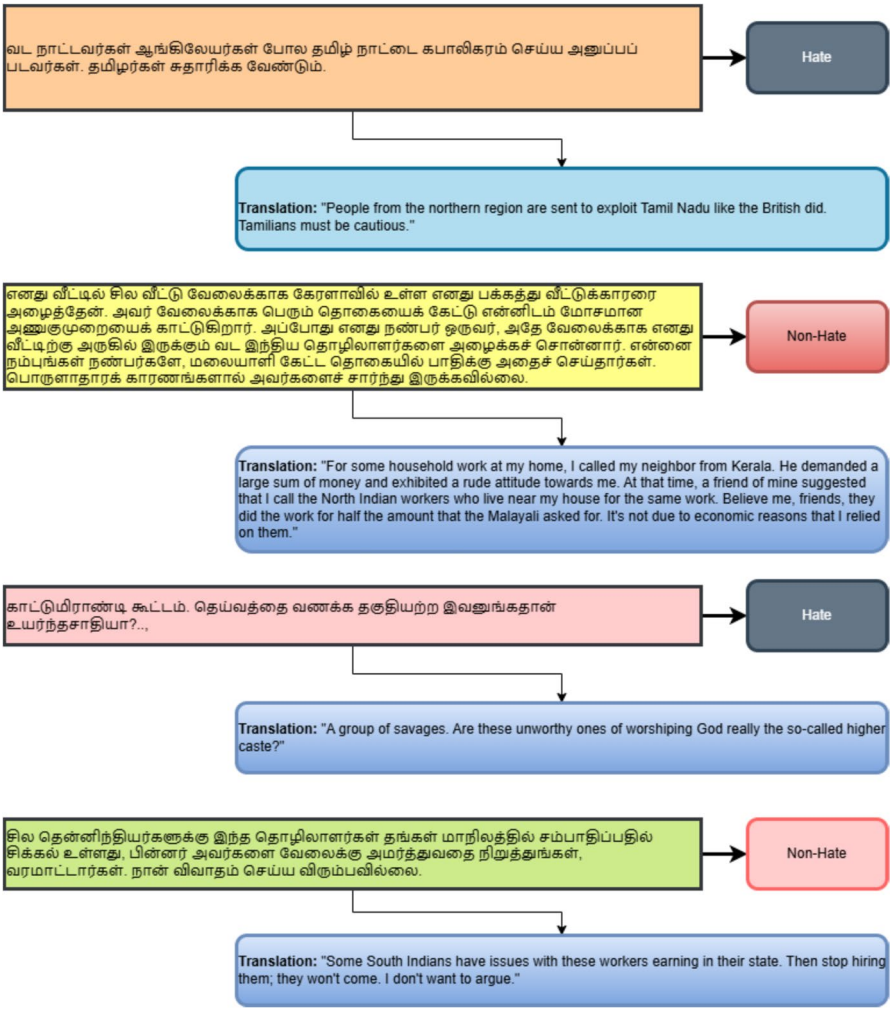


Fig. 4 The sample annotated comments from dataset with the translation. The label "Hate" means the hate speech related to caste/migration, and "Not-Hate" means not hate speech related to caste/migration

The bidirectional nature of sentences is effectively captured through this approach, allowing the model to learn contextual representations that account for both preceding and following words in the input sequence.

XMOD: X-MOD (Pfeiffer et al., 2022) is a multilingual masked language model that has been trained on filtered CommonCrawl data, which includes content from 81 different languages. Unlike previous multilingual models such as XLM-R (Conneau et al., 2020), X-MOD introduces a novel approach by incorporating language-specific modular components known as language adapters during its pre-training phase. In X-MOD, these language adapters are frozen in each transformer layer during fine-tuning, enabling efficient adaptation to

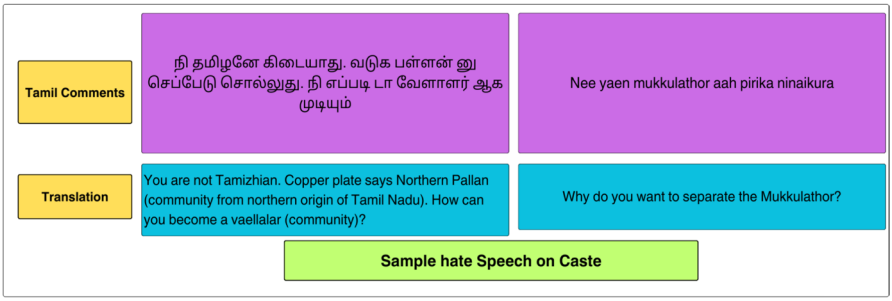


Fig. 5 The sample comment related to caste hate speech from the dataset. The text inside the purple box is the comment and the blue box is the translation

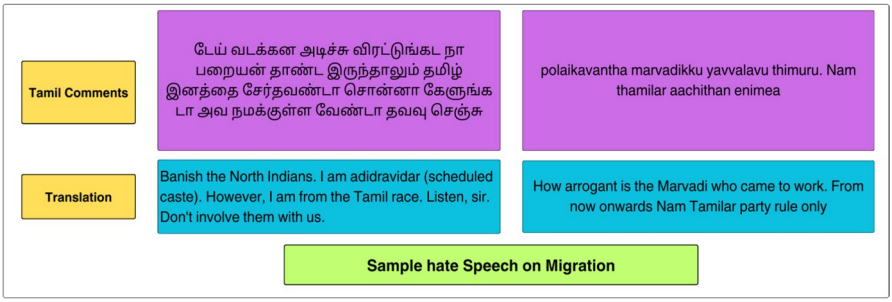


Fig. 6 The sample comment related to migration hate speech from the dataset. The text inside the purple box is the comment and the blue box is the translation

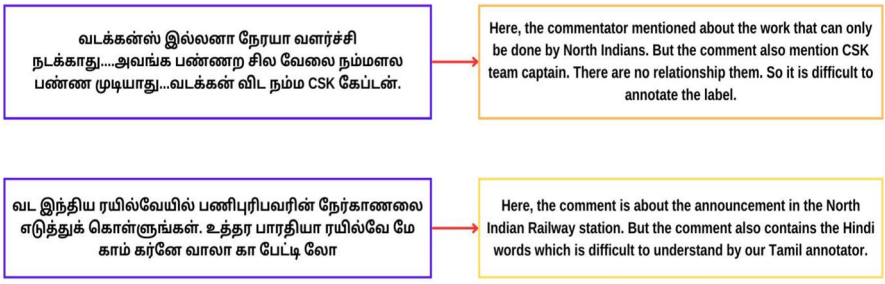


Fig. 7 The sample comment from the dataset, which we found difficult to annotate is outlined in purple box. The yellow box contains explanation of the difficulties encountered during an annotation process

specific languages without modifying the underlying model architecture. Moreover, X-MOD utilizes the same tokenizer as XLM-R, ensuring compatibility and leveraging existing infrastructure for tokenization across languages. This design

Table 3 Specifications of the experimental platform used for model training and evaluation

Item name	Specifications
GPU	Nvidia Tesla V100
RAM	156 GiB
GPU RAM	16 GiB
OS	Ubuntu
Language	Python
IDE	VS Code editor

choice facilitates seamless integration into existing frameworks and workflows that are based on XLM-R or similar models.

4.1.1 Default loss

All models were trained using 5 different learning rates over 5 epochs, alongside other specified hyperparameters. The objective was to determine the optimal learning rate that maximizes performance for each model. The selected learning rates for the experiment were: 1.00E−05, 2.00E−05, 3.00E−05, 4.00E−05, and 5.00E−05. The learning rate yielding the highest macro F1-score was chosen as the best for each model.

Following the evaluation of baseline models, further experiments were conducted to enhance their performance. These experiments involved the implementation of custom loss functions, adapter modules, and parameter-efficient fine-tuning strategies. The objective was to optimize model refinement and improve their effectiveness in specific tasks or domains.

4.1.2 Custom loss

The three following loss functions were chosen for the experiments.

1. *KL divergence Loss:* The Kullback–Leibler (KL) divergence, denoted as $KL(P||Q)$, is a statistical measure that quantifies how one probability distribution P diverges from another reference distribution Q. In the context of evaluating model outputs against true labels, P typically represents the predicted distribution while Q represents the actual distribution of the labels.

   A higher KL divergence score indicates greater dissimilarity between the two distributions P and Q. Conversely, a KL divergence score of zero signifies that the distributions P and Q are identical.

$$KL(y_c||\hat{y}_c) = \sum_{c=1}^{M} \hat{y}_c \log \frac{\hat{y}_c}{y_c} \tag{2}$$

   where y_c = True labels, $\hat{y}_c$ = Predictions.

2. *Hinge Loss:* The hinge loss is another popular loss function used for training classifiers. The hinge loss of the prediction y is defined as:

$$HL(y) = max(0, 1 - t \cdot y) \tag{3}$$

Where, $t = \pm 1$ and y is the "raw" output of the classifier's decision function

3. *Binary Cross Entropy Loss:* Binary Cross Entropy is the negative average of the log of corrected predicted probabilities.

$$BCE = -[y * \log(p(\hat{y})) + (1 - y) * \log(1 - p(\hat{y}))]) \tag{4}$$

where y = True labels, $p(\hat{y})$ = Probability of predicted label.

4.2 Adapter based fine-tuning

In each layer of a Transformer model, Bottleneck adapters (Hu et al., 2023) introduce bottleneck feed-forward layers characterized by a *W_down* matrix for down-projection, reducing the dimensionality *d_bottleneck* of layer hidden states, an activation function f, *W_up* for up-projection restoring the original dimension, and a residual connection r.

$$h \leftarrow W_{up} \cdot f(W_{down} \cdot h) + r \tag{5}$$

These layers can be incorporated at various positions within a Transformer block based on the adapter configuration. Further customization is possible for layer norms, activation functions, residual connections, and bottleneck sizes. The bottleneck dimension *d_bottleneck* stands out as a crucial configuration hyperparameter. It is indirectly defined through the *reduction_factor* attribute in the configuration, representing the ratio between a model's layer hidden dimension and the bottleneck dimension, i.e.:

$$reduction_factor = \frac{d_{hidden}}{d_{bottleneck}} \tag{6}$$

Figure 8 provides a visual representation of additional configuration options pertaining to the adapter structure. This adapter is implemented with the help of "BnConfig" from AdapterHub[24] and LLM-Adapters package (Hu et al., 2023).

4.3 Parameter-efficient fine-tuning

Parameter-efficient fine-tuning (PEFT)[25] is a technique employed in NLP to enhance the performance of pre-trained language models when applied to specific downstream tasks. Unlike traditional fine-tuning where all layers and parameters of the pre-trained model are adjusted, PEFT selectively freezes certain layers and focuses

[24] https://docs.adapterhub.ml/methods.html.
[25] https://huggingface.co/docs/peft/index.

solely on fine-tuning the last few layers relevant to the new task. This strategy conserves computational resources and time compared to full fine-tuning from scratch. Moreover, PEFT enables effective adaptation of the model to new tasks even when limited labeled data is available. By targeting only the essential parameters for the task at hand, PEFT optimizes efficiency without compromising performance, making it a practical choice for scaling up models in resource-constrained environments.

4.3.1 P-tuning

In P-tuning,[26] the traditional discrete prompt is augmented with trainable continuous prompt embeddings, which serve as additional learnable parameters. Unlike manually designed text prompts, these continuous embeddings are optimized via backpropagation, enhancing both stability and effectiveness in steering a pre-trained language model (LM). Figure 9 illustrates the workflow of the P-Tuning mechanism. Specifically, let M denote a pre-trained language model with a hidden size h and a vocabulary size V. Consider a labeled dataset (x_i, y_i) for an NLU task, where $x_{0:n} = \{x_0, x_1, \ldots, x_n\}$ forms a sequence of discrete tokens constituting the input, and $y \in Y$ represents a label. The objective is to estimate the conditional probability $f_M(x) = \hat{p}(y|x)$, with M's parameters either fine-tuned or frozen. Recognizing that discrete prompts often exhibit high instability and suboptimal behavior under backpropagation, P-Tuning introduces continuous prompt embeddings to enhance stability and effectiveness in prompting.

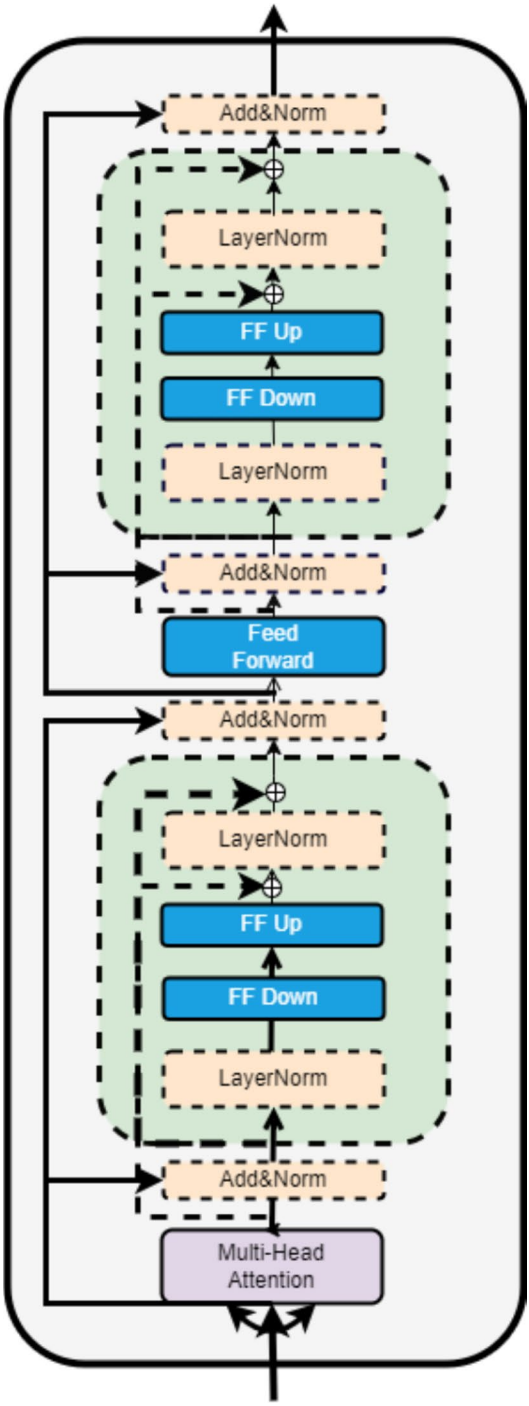
The prompt template used in P-Tuning is structured as follows: $T = \left\{\left[P_{0:i}\right], x, \left[P_{(i+1):j}\right], y, \left[P_{(j+1):k}\right]\right\}$, where each $\left[P_i\right]$ represents the (ith) continuous prompt embedding. P-Tuning incorporates an additional embedding function $f : \left[P_i\right] \rightarrow h_i$, which maps the template to $h_0, \ldots, h_i, e(x), h_{i+1}, \ldots, h_j, e(y), h_{j+1}, \ldots, h_k$. These embeddings are then updated, $\left[P_i\right]_{i=1}^{k}$, to optimize a task-specific loss function. Importantly, P-Tuning is applicable to both frozen and fine-tuned language models, demonstrating versatility across different model configurations.

4.3.2 LoRA

LoRA,[27] or Low-Rank Adaptation of large language models (Hu et al., 2022), is a technique in PEFT aimed at reducing the number of trainable parameters. It achieves this by integrating trainable rank decomposition matrices into each layer of the Transformer architecture, thereby decreasing the number of trainable parameters for subsequent tasks. It's crucial to emphasize that the parameters of the original pre-trained transformer remain static. Aghajanyan et al. demonstrate that over-parametrized models learned on a low intrinsic dimension support the hypothesis that weight changes during model adaptation also possess a low "intrinsic rank", forming the foundational

[26] https://huggingface.co/docs/peft/package_reference/p_tuning.
[27] https://huggingface.co/docs/peft/package_reference/lora.

Fig. 8 Adapter architecture



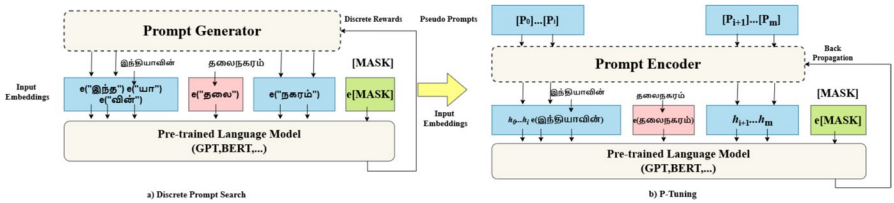


Fig. 9 Working diagram of P-Tuning

principle of LoRA. The study additionally asserts that LoRA enables the indirect training of select dense layers within a neural network by optimizing rank decomposition matrices of these layers' adaptations, all while maintaining the frozen state of pre-trained weights.

Figure 10 shows the work flow of LoRA. "d" represents the number of input features and "h" represents the number of output features, and they are connected through the weight matrix that transforms the 'd'-dimensional input into an 'h'-dimensional output. A pre-trained weight matrix 'W_0' is expressed using a low-rank decomposition to constrain its updates:

$$W_0 + \Delta W = W_0 + BA \tag{7}$$

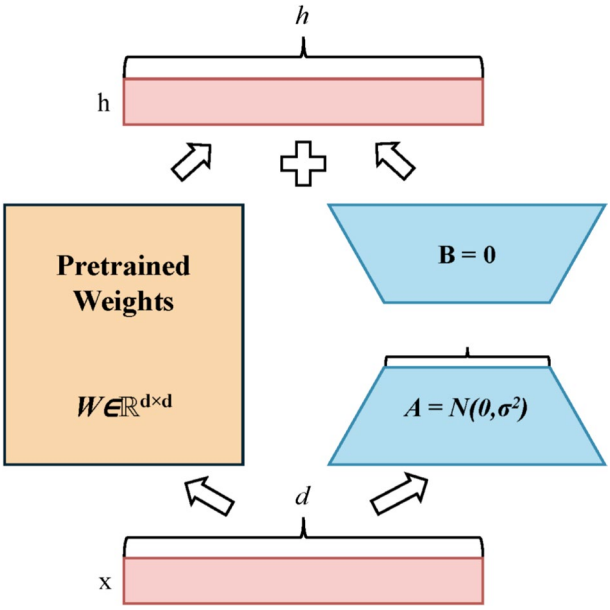


Fig. 10 Working diagram of LoRA technique. In this figure, 'x' represents the number of input features, where '*d*' denotes the dimensionality of x. Similarly, 'h' represents the number of output features, where '*h*' corresponds to the dimension of the transformed input

During training, W_0 remains frozen and does not receive gradient updates, while A and B consist of trainable parameters. For $h = W_0x$, the modified forward pass is given by:

$$h = W_0x + \Delta Wx = W_0x + BAx \tag{8}$$

Initially, A is randomly initialized using a Gaussian distribution, and B is initialized to zero, ensuring that $\Delta W = BA$ is zero at the start of training. In the Transformer architecture, there are four weight matrices in the self-attention module (W_q, W_k, W_v, W_o) and two in the MLP module. LoRA adapts only the attention weights for downstream tasks while freezing the MLP modules. In summary, the LoRA technique decomposes the weight matrix into low-rank matrices that are trained and updated, keeping all the pre-trained model parameters frozen. After training, the low-rank matrices are incorporated back into the original weights. This approach simplifies training and storage of LoRA models due to the significantly reduced number of parameters.

5 Results

The performance of our models was based on the macro-F1 metric, calculated using the scikit-learn[28] library. This metric is used because it provides insights into both the overall effectiveness of the models and their ability to manage class imbalances, which is particularly challenging in the context of caste and migration-related hate speech detection. In this study, the efficacy of the language models was assessed using a dedicated test set. Tables 4 and 5 shows the results of the baseline and the custom methods.

A classification layer was appended to the pre-trained models to fine-tune the models using the training set for the baseline results. The performance of these fine-tuned models on the test set varied significantly based on the learning rates and pre-trained weights used. Among the tested models, TamilSBERT demonstrated the highest performance with a macro-F1 score of 0.70 at a learning rate of 2.00E−05, highlighting its effectiveness in handling the task. Other competitive models included Bloom-560 m, Muril-base-cased, and xmod-base, each achieving a score of 0.69 with optimal learning rates of 4.00E−05, 2.00E−05, and 3.00E−05, respectively. Tamil-BERT followed closely with 0.67 at 1.00E−05, while IndicBERT achieved 0.63 at the same learning rate. In contrast, XLM-RoBERTa base underperformed, scoring only 0.50 at its best learning rate of 1.00E−05, suggesting its limitations in this specific task. These results indicate TamilSBERT's superior contextual understanding, while XLM-RoBERTa and IndicBERT may require further optimization for better performance.

Next, we explored custom models by modifying loss functions and integrating adapter and PEFT techniques, using the optimal learning rates identified in the

[28] https://scikit-learn.org/stable/.

baseline experiments to enhance model performance. The results varied significantly based on the techniques applied. Among the loss functions tested, KL loss proved the most effective, enabling Bloom-560 m to achieve the highest macro-F1 score of 0.73, marking a substantial improvement over its baseline. BCE loss also contributed to incremental gains, improving TamilBERT by 2% (0.69), TamilSBERT by 1% (0.69), and significantly boosting XLM-RoBERTa by 18% (0.68). In contrast, Hinge loss consistently yielded poor performance (0.22–0.47) across all models, indicating its unsuitability for this task. For adapter-based techniques, the bottleneck adapter with XMOD reached 0.59, though it remained below the fine-tuning results. The LoRA (PEFT) technique improved XMOD to 0.62, but still fell short of KL loss, which performed 7% better. Similarly, P-tuning (PEFT) with Bloom-560 m achieved a macro-F1 score of 0.51, showing limited improvements.

Overall, Bloom-560 m fine-tuned with KL loss at a learning rate of 4.00E−05 outperformed all other models and methods, achieving the best macro-F1 score of 0.73. This superior performance can be attributed to the KL loss function's effectiveness in handling the nuances of caste and migration-related hate speech detection. The KL loss function likely helped the model better differentiate between subtle variations in the text, which is crucial for accurately identifying hate speech. This result underscores the importance of selecting an appropriate loss function to optimize model performance in complex NLP tasks. These findings indicate that choosing the right loss function has a greater impact on model performance than using adapter-based or PEFT techniques, as even the best adapter-based methods fell short of fine-tuning with KL loss. Furthermore, while BCE loss helped certain models, its improvements were marginal compared to KL loss, emphasizing that loss function choice plays a pivotal role in improving classification accuracy for nuanced text-based tasks.

6 Error analysis

Error analysis is crucial in understanding the limitations of our classification model and identifying patterns in misclassification. Despite achieving high performance, the BLOOM with KL loss model occasionally misclassifies comments due to various linguistic complexities, including metaphorical expressions, context sensitivity, and reliance on specific keywords. In this section, we analyze five misclassified examples to highlight possible reasons for incorrect predictions and suggest improvements. Figure 11 presents the five misclassified examples along with their transliteration and translation, aiding in a deeper understanding of the errors. The misclassifications fall into several categories, including keyword-based misclassification, failure to recognize indirect hate speech, challenges with Romanized Tamil, difficulties with informal and dialectal variations, and over-sensitivity to identity-based discussions.

One such case is Comment ID 50, where the model misclassified a non-hate speech comment as hate speech. The phrase "Karuppu" (black) was misinterpreted as a racial or casteist term, though in this context, it metaphorically referred to "black sheep" within the Tamil community. The error stemmed from

Table 4 Baseline results with selective learning rates with various LLMs assessed with macro-F1 score

Learning rates	BLOOM	MuRIL	TamilBERT	TamilSBert	XLM-RoBERTa	XMOD	IndicBERT
1.00E−05	0.63	0.65	**0.67**	0.68	**0.50**	0.67	0.62
2.00E−05	0.54	**0.69**	0.66	**0.70**	0.38	0.67	0.62
3.00E−05	0.65	0.38	0.64	0.69	0.38	**0.69**	**0.63**
4.00E−05	**0.69**	0.38	0.64	0.38	0.38	0.38	0.38
5.00E−05	0.54	0.38	0.38	0.57	0.38	0.38	0.38

Highest values are highlighed in bold

Table 5 Results of the custom models with the modification of custom loss, Adapter and PEFT techniques assessed with macro-F1 score

Methods	BLOOM	MuRIL	TamilBERT	TamilSBert	XLM-RoBERTa	XMOD	IndicBERT
	4.00E−05	2.00E−05	1.00E−05	2.00E−05	1.00E−05	3.00E−05	3.00E−05
Default loss	0.69	**0.69**	0.67	0.68	0.50	**0.69**	**0.63**
BCE loss	0.42	0.38	**0.69**	**0.69**	**0.68**	0.38	0.38
KL loss	**0.73**	0.38	0.38	0.38	0.38	0.38	0.38
Hinge loss	0.22	0.28	0.28	0.28	0.28	0.28	0.28
Bottleneck adapter	0.47	0.38	0.38	0.39	0.39	0.59	0.40
LoRA (PEFT)	0.58	0.38	0.38	0.38	0.44	0.62	0.38
P-tuning (PEFT)	0.51	0.38	0.38	0.38	0.38	0.38	0.38

Highest values are highlighed in bold

keyword-based misclassification, where the system over-relied on specific words rather than understanding their contextual meaning. Similarly, Comment ID 82 expressed concerns about financial exploitation, suggesting that an unnamed group benefits at the expense of Tamilians. Though the comment implicitly reinforces an "Us vs. Them" narrative-often present in migration-related hate speech-the model misclassified it as non-hate speech. This error highlights the model's failure to detect implicit bias and exclusionary rhetoric, likely due to its dependence on explicit identity markers. In Comment ID 96, the model misclassified a comment targeting the Marwari (Marvadi—people originating from the Marwar region of Rajasthan, India) community as non-hate speech. A key reason for this misclassification was Romanized Tamil confusion-since the comment was written in English letters instead of Tamil script, the model struggled to correctly interpret its meaning. Additionally, informal Tamil and slang usage further complicated detection, as the phrase "thimir pidichavanugha" (arrogant people) is a colloquial expression that may not have been well-represented in the training data.

Further, Comment ID 136 blended formal and informal Tamil, making it harder for the model to classify it correctly. The statement promoted exclusionary political rhetoric, arguing that jobs should be reserved exclusively for Tamil

speakers. The model misclassified this as non-hate speech due to difficulties in analyzing mixed-style language and informal dialects, as well as challenges in recognizing coded exclusionary speech. Similarly, Comment ID 121 was wrongly classified as hate speech due to over-sensitivity to identity-based statements. The comment discussed government job allocations, arguing that non-Tamil officials prioritize hiring individuals from their own states. While this could contribute to regional bias, the comment itself lacks explicit slurs or aggressive language, making it more of a policy concern than hate speech. The model's inability to differentiate between political/economic complaints and actual hate speech likely led to this false positive.

To address these misclassifications in future, the model must be enhanced with deeper contextual understanding, allowing it to analyze full sentence meanings rather than relying on specific keywords. Metaphor recognition should be improved to distinguish between figurative language and explicit hate speech, preventing errors such as the one in Comment ID 50. Additionally, self-criticism differentiation should be refined to prevent misclassifying intra-community criticism as hate speech. Recognition of indirect hate speech must be strengthened to detect implicit biases and exclusionary rhetoric, even when explicit identity markers are absent, which would help resolve errors like those in Comment ID 82. Furthermore, the model must recognize "Us vs. Them" narratives and be trained on Romanized Tamil, slang, and informal speech to better classify hate speech written in mixed-language contexts. Coded exclusionary speech detection should be enhanced to capture political rhetoric that indirectly promotes discrimination while ensuring the model can distinguish policy discussions from actual hate speech. Finally, the model should reduce over-sensitivity to identity-related discussions, ensuring that factual demographic statements are not unnecessarily flagged. Implementing these refinements will significantly enhance the system's ability to detect nuanced hate speech while reducing both false positives and false negatives.

7 Shared task overview

7.1 Task description

A shared task named "Caste and Migration Hate Speech Detection" (Rajiakodi et al., 2024) was conducted as part of LT-EDI@EACL 2024 with the dataset created. The main objective of the task is to create an automatic classification system that can precisely detect the occurrence of hate speech related to caste/migration in textual content. This shared task is conducted via the Codalab competition.[29] A total of 51 teams participated, out of which 15 teams submitted their entries. The participants were provided with training, development, and test datasets. As far as we know, this is the first collaborative effort focused on identifying hate speech related to caste and

[29] https://codalab.lisn.upsaclay.fr/competitions/16089.

migration in social media comments. The annotation has been categorized into two groups: Hate speech on caste/migration or Not-Hate speech on caste/migration.

7.2 Participants methodology and results

The methodologies used by the participants are described below, and the rank list is shown in the Table 6.

- *BITS_GraphAI:* This team employed two pre-trained transformers, namely TamilSBERT-STS and Indic-SBERT, as described in the works of Deode et al. (2023) and Mirashi et al. (2023) respectively. They then fine-tuned these transformers using the provided data, utilizing triplet and cosine similarity loss functions, respectively. Triplet loss involves using triples in the format of (anchor, positive, negative), where the anchor and positive sentences belong to the same class, while the negatives are taken from other classes. Triplet loss enhances the model's ability to distinguish subtle correlations. In order to improve the classification process, they employed the TextGCN architecture (Yao et al., 2019) by utilizing the fine-tuned sBERT embeddings as feature input and the text graph as an adjacency matrix. All three models exhibited strong performance, with sBERT-enhanced TextGCN graph-based learning demonstrating slightly superior outcomes. The team attained a ranking of 4, accompanied by a macro F1 score of 0.77.
- *SSN-Nova* (Reddy et al., 2024): This team explores various boosting techniques, including Adaboost and XGBoost (Demir & Sahin, 2023). They also conduct a comparative analysis using a voting classifier that combines multiple traditional models. This approach thoroughly investigates ensemble methods, utilizing the advantages of both boosting algorithms and conventional models to improve the system's predictive performance. They achieved a macro F1 score of 0.59, securing the 12th rank in the process.
- **Kubapok** (Pokrywka & Jassem, 2024): The team adopted a systematic approach, relying exclusively on the data provided by the organizers. Their methodology focused on using a variety of models, including 'l3cube/pune-kannada' (Deode et al., 2023; Mirashi et al., 2023), 'microsoft-mdeberta-v3' (He et al., 2021) (used twice), and 'XLM-RoBERTa'. These models were trained using standard Hugging Face scripts for text classification, incorporating adjustments like a warm-up ratio of 0.1 and running for 30 epochs. Importantly, they combined the training and development data, creating new random splits for each model. The optimal epoch checkpoint was selected based on the development F1 score. A critical part of their strategy was averaging the probabilities generated by all four models and applying a threshold of 0.5 for class selection. This approach led the team to achieve rank 2 with a macro F1 score of 0.81.
- *KEC_AI_DSNLP* (Shanmugavadivel et al., 2024): This team developed machine learning models including K-Nearest Neighbors, Decision Trees, and Naive Bayes to classify hate speech text. Their approach yielded a macro F1 score of 0.65, securing them the 9th rank.

Table 6 Participants rank list of the shared task based on average macro F1 Score

Participant's team names	Macro F1	Rank
Transformers_run3 (Singhal & Bedi, 2024)	0.82	1
kubapok_run1 (Pokrywka & Jassem, 2024)	0.81	2
CUET_NLP_Manning_run3 (Alam et al., 2024)	0.80	3
BITS_Graph4NLP_run1	0.77	4
Algorithmalliance_run1 (Sangeetham et al., 2024)	0.76	5
lidoma_run2 (Tash et al., 2024)	0.76	6
CUET_NLP_GoodFellows_run2	0.75	7
quartet_run1 (Shaun et al., 2024)	0.73	8
KEC_AI_DSNLP_run2 (Shanmugavadivel et al., 2024)	0.65	9
selam_run1	0.62	10
byteSizedllm_run1	0.61	11
SSN-nova_run3 (Reddy et al., 2024)	0.59	12
WordWizards_tamil_run1	0.54	13
KEC_DL_KSK_run2	0.49	14
Habesha_run1	0.38	15

Comment_id	Comment	Original_label	Predicted_label
50	தமிழனுக்கு தமிழன் தான் எதிரி தமிழ்நாட்டிலேயே பல கருப்பு அடங்கியிருக்கிறது அப்ப எப்படி தமிழ் மக்கள் வாழ முடியும் **Transliteration:** Tamizhanukku tamizhan dhaan edhiri tamizhilnaattileye pala karuppu adangirukkuradhu appa yeppidi Tamiil makkal vazha mudium **Translation:** A Tamizhian is an enemy to another Tamizhian. There are many dark issues hidden within Tamil Nadu itself. Then how can Tamil people live?	Non-Hate	Hate
82	நம்மிடம் வட்டி வாங்கி கொழுத்து நம் பணத்தை வைத்தே நம்மை அடிமைபடுத்தி விடுவார்கள் இவர்கள் கண்காணிக்க படவேண்டியவர்கள் தமிழர்கள் இனியாவது எச்ரிக்கையுடன் இருக்க வேண்டும் **Transliteration:** Nammidam vatti vaangi kozhuttu, nam panathai vaitthe nammai adimaipaduthi viduvaargal. Ivargal kankaanikka padavendiyavargal. Tamizhargal iniyavathu echarikkaiyudan irukka vendum. **Translation:** They will take interest from us and grow rich, then use our own money to enslave us. These people need to be watched. Tamilians must at least now be cautious.	Hate	Non-Hate
96	Marvadi yellarum ipdi dha thimir pidichavanugha **Translation:** All Marvadis are like this - arrogant people.	Hate	Non-Hate
136	Tamilaga makkalai kurai sollade kattadavela pakradlendu Yella velayum pakran tamilnadu tamil nada irukanumna tamilanukku matum Vela kudu basa puriyada aalta nee Inna Vela vangi kilikka pora innada pesriya ollu okra ipadikku nam tamilan seemanin vilutugal **Translation:** Don't find fault with Tamil people. People from other states are doing construction work for others in Tamil Nadu. If you want Tamil Nadu to remain Tamil Nadu, give the work to Tamil people. What is the point of giving jobs to those who don't even understand the language? – Thus speak the followers of 'Nam Tamilar' Seeman.	Hate	Non-Hate
141	Neenga vadakkan makkalai paarkaathirgal.....in all education department , collectors , government officers are all other state....so they make entered those state people. It's very simple. Maela avana ukkaaara vaechutta. Avan avan aaalungala keel thazhathilum ukkaaara vaechutaaan. **Translation:** You are only noticing the northern people... In all education departments, collectors, and government offices, it's mostly people from other states. So, they bring in more people from their own states. It's very simple - once you place them in power, they, in turn, place their own people in lower positions as well.	Non-Hate	Hate

Fig. 11 Displayed the five misclassified comments analyzed in the error analysis section, showing the original and predicted labels

- *CUET_NLP_GoodFellows:* This team utilized two BERT models, specifically mBERT and XLM-R, both of which were fine-tuned for their task. In addition to these, they also employed a fine-tuned random classifier.
- *selam:* This team employed a Support Vector Machine (SVM) approach within the realm of Natural Language Processing (NLP). The primary goal was to develop a robust text classification system capable of predicting whether a given text contains caste/migration hate speech and scored macro F1 of 0.62 and secured rank 10.
- *KEC_DL_KSK:* This team utilized sampling techniques such as SMOTE and Random Oversampling to balance the datasets. They implemented machine learning algorithms, including Random Forest, Support Vector Machine, Naive Bayes, Logistic Regression, and Decision Tree. Additionally, they employed word embedding methods such as TF-IDF, Word2Vec, Doc2Vec, and FastText. Their approach resulted in a macro F1 score of 0.49.
- *bytesizedllm:* This team utilized embeddings derived from a subset of AI4B-harat's data, consisting of 100,000 randomly selected lines. These embeddings were generated using their custom-built subword tokenizers for Telugu (7.6 MB) and Tamil (1.3 MB). They employed a Bidirectional Long Short-Term Memory (BiLSTM) classifier for classification tasks. The model was trained on labeled datasets and achieved a macro F1 score of 0.61.
- *Word Wizards:* This team utilized LaBSE (Pei et al., 2022), a pre-trained language representation model designed for multilingual text understanding. LaBSE employs a Siamese encoder architecture, which generates high-quality sentence embeddings by encoding text into fixed-size vector representations. After obtaining these embeddings, a K-Nearest Neighbors (KNN) model was implemented, likely for tasks such as similarity searches or classification based on the encoded representations. KNN, a simple yet effective algorithm, is widely used for classification and regression, especially in high-dimensional spaces. This combination of LaBSE embeddings and KNN likely facilitated tasks involving semantic similarity, clustering, or text classification. The team secured 13th place with a macro F1 score of 0.54.
- *Lidoma* Tash et al. (2024): This team utilized both deep learning and machine learning models, including Convolutional Neural Networks (CNN) and Support Vector Machines (SVM), to classify hate speech related to caste and migration in the Tamil language. Their approach achieved a macro F1 score of 0.76, securing them the 6th rank in the competition.
- *Transformers* Singhal and Bedi (2024): This team employed an ensemble model consisting of XLM-RoBERTa, a multilingual BERT-based model, and the MuRIL cased model (Subramanian et al., 2022). These models achieved the highest accuracy compared to all other tested models. The ensemble approach further enhanced overall accuracy by leveraging the strengths of both models. Notably, all models were trained on raw text without preprocessing, as data cleaning negatively impacted their performance.
- *CUET_NLP_Manning* (Alam et al., 2024): This team implemented six machine learning models (Logistic Regression, Support Vector Machine, Stochastic Gradient Descent, XGBoost, Ensemble, and Random Forest), three deep learning mod-

els (BiLSTM, Attention, and BiLSTM-CNN), and three transformer-based models (M-BERT, XLM-R, and Tamil-BERT) using TF-IDF and FastText embeddings. Among these, the transformer-based models demonstrated the best performance, achieving the highest evaluation scores. The team secured 3rd place with a macro F1 score of 0.80.

- *Habesha:* This team developed a model based on BERT transformers for generating embeddings. Additionally, they incorporated Long Short-Term Memory (LSTM) deep learning techniques to enhance classification tasks. This hybrid architecture effectively utilizes transformer-based embeddings for rich text representation while leveraging LSTM's sequential learning capabilities for improved classification. Their approach resulted in a macro F1 score of 0.38.
- *ALGORITHM ALLIANCE* (Sangeetham et al., 2024): This team implemented various supervised machine learning algorithms, including Logistic Regression, (SVM), Multi-Layer Perceptron (MLP), Random Forest Classifier (RFC), Decision Tree, and K-Nearest Neighbors (KNN) for classification tasks. Among these models, SVM achieved the highest performance with a macro F1 score of 0.76, securing the 5th rank in the competition.
- *Quartet* (Shaun et al., 2024): This team employed machine learning models, including Logistic Regression, SVM, Random Forest Classifier, Decision Tree Classifier, and Naive Bayes, for classification tasks. They used TF-IDF for feature representation. Among these models, SVM demonstrated the best performance, achieving a macro F1 score of 0.73 and securing the 8th rank in the competition.

8 Conclusion

In this study, we addressed the significant challenge of detecting caste and migration-related hate speech in Tamil language by developing a specialized dataset and leveraging advanced multilingual models. Our experiments revealed that the choice of model and fine-tuning strategies greatly influence performance, with tamil-sentence-bert-nli achieving the highest baseline score of 0.69. We further enhanced model performance by experimenting with various loss functions and adapter techniques. Notably, Bloom-560 m combined with the KL loss function reached an impressive macro-F1 score of 0.73, highlighting the importance of loss function selection in capturing nuanced hate speech patterns. These findings underscore the potential of leveraging state-of-the-art NLP techniques and thoughtful experimental design to tackle hate speech in low-resource languages, paving the way for more effective and culturally aware automated moderation systems. Future work will focus on refining these models and exploring additional methods to further enhance detection accuracy and robustness.

9 Ethical considerations

In conducting our study, which involves analyzing YouTube comments, we have prioritized ethical considerations throughout the data collection process. We took significant measures to safeguard the privacy of commentators, ensuring that their

identities were protected and that the comments were used responsibly and in a manner that would not cause harm. Moreover, we strictly limited the distribution of data to academic and non-commercial research purposes only, thereby adhering to ethical standards that prioritize privacy and prevent misuse.

By conducting our research ethically and responsibly, we aimed to minimize potential harm, uphold privacy protections, and contribute meaningfully to our field of study. These efforts underscore our commitment to conducting research that is both rigorous and respectful of ethical principles. In conducting this study, which involves the utilization of YouTube comments, we have taken into account many ethical implications while collecting the YouTube comments we made sure that the privacy of commentators was well protected and that the comments were not used in a way that could have caused any harm. In addition, we ensured that data distribution is prohibited for anything but academic and non-commercial research uses. Thus we have done our research ethically and responsibly to reduce harm, protect privacy, and make meaningful contributions to the field of study.

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Data availability Data is provided in shared task https://codalab.lisn.upsaclay.fr/competitions/16089.

Declarations

Conflict of interest The authors declare no conflict of interest.

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Authors and Affiliations

Bharathi Raja Chakravarthi[1,3] · Saranya Rajiakodi[2] · Rahul Ponnusamy[3] · Bhuvaneswari Sivagnanam[2] · Sara Yogesh Thakare[4] · Sathiyaraj Thangasamy[5]

✉ Bharathi Raja Chakravarthi
bharathi.raja@universityofgalway.ie

Saranya Rajiakodi
saranya@cutn.ac.in

Rahul Ponnusamy
r.ponnusamy1@universityofgalway.ie

Bhuvaneswari Sivagnanam
bhuvisiva24@gmail.com

Sara Yogesh Thakare
sara.thakare.22033@iitgoa.ac.in

Sathiyaraj Thangasamy
sathiyarajkkt@gmail.com

1 School of Computer Science, University of Galway, Galway, Ireland

2 Department of Computer Science, Central University of Tamil Nadu, Thiruvarur, Tamil Nadu, India

3 Data Science Institute, University of Galway, Galway, Ireland

4 Indian Institute of Technology Goa, Bandiwada, India

5 Department of Tamil, Sri Krishna Adithya College of Arts and Science, Kovaipudur, Tamil Nadu, India